

Activities: Day 1

1) Find 10 different model providers and their models released in last 12 months

Details to collect:

Model Provider | Model | Release Date | Cut off Date | Cost of tokens | Context Window | Type |
When to use this model |

Solutions:

Model Provider	Model	Release Date	Knowledge Cutoff	Cost of Tokens	Context Window	Type	When to use this model
<u>OpenAI</u>	GPT-5.5	Apr 23, 2026	Dec 1, 2025	Input \$5 / Output \$30 ; cached input \$0.50	1.05M	Frontier reasoning / coding	Complex professional work, advanced coding, computer-use agents, research, difficult reasoning.
<u>Anthropic</u>	Claude Opus 4.8	May 28, 2026	Jan 2026	Input \$5 / Output \$25 ; cached input \$0.50	1M	Hybrid reasoning / agentic	Premium coding, long-running AI agents, complex knowledge work, enterprise documents and high-accuracy tasks.
<u>Google DeepMind / Gemini</u>	Gemini 3.1 Pro	Feb 19, 2026	Jan 2025	Input \$2 / Output \$12 (<200K input); \$4/\$18 above 200K	1.05M	Multimodal reasoning	Complex multimodal analysis, large documents/code bases, advanced reasoning, software engineering and agentic workflows.
<u>xAI</u>	Grok 4.1 Fast	Nov 19, 2025	Not publicly disclosed	Input \$0.20 / Output \$0.50 ;	2M	Fast reasoning / agentic	Cost-efficient agents, tool calling, web/X search, customer

				cached input \$0.05			support, finance and real-time research.
<u>Mistral AI</u>	Mistral Small 4	Mar 16, 2026	Not publicly disclosed	Input \$0.15 / Output \$0.60	256K	Hybrid instruct + reasoning + coding	High-volume production, coding, reasoning, function calling and agents where low cost and open weights matter.
<u>Cohere</u>	Command A+	May 20, 2026	Apr 1, 2025	Free until rate limits on Cohere trial/production keys; Model Vault for production	128K	Enterprise MoE / multimodal reasoning	Enterprise agents, RAG, multilingual applications, document processing, sovereign/private deployments and multimodal workflows.
<u>Alibaba / Qwen</u>	Qwen3-Coder-480B-A35B-Instruct	Jul 22, 2025	Jun 2025	Alibaba Global: roughly \$1–\$6 input / \$5–\$60 output depending on context tier; third-party routed rates can be lower	256K native / 1M extended	Open-weight MoE coding	Large codebases, autonomous coding agents, repository analysis, debugging, tool use and long-context software engineering.
<u>Z.ai</u>	GLM-5	Feb 12, 2026	Not publicly disclosed	About \$1.00 input / \$3.20 output ; cached input ~\$0.20	200K	Open-weight reasoning / agentic coding	Complex systems engineering, long-horizon coding agents, autonomous tool use and self-hosted/open-weight deployments.

<u>DeepSeek</u>	DeepSeek-V4-Pro	Apr 24, 2026	Not publicly disclosed	Input \$0.435 / Output \$0.87 ; cached input \$0.003625	1M	Open-weight reasoning / agentic	Very cost-sensitive reasoning, coding, agents, long documents and workloads requiring a huge context window.
<u>Moons</u> <u>hot AI</u> / <u>Kimi</u>	Kimi K3	Jul 16, 2026	Not publicly disclosed	Input \$3 / Output \$15 ; cached input \$0.30	1.05M	Large MoE / multimodal reasoning	Long-horizon coding, autonomous agents, large knowledge-work tasks, very long documents and multimodal workflows.

2) Read 2–3 articles about how transformer works?

Goal to understand:

- (a) Self-attention mechanism
- (b) Vector embedding
- (c) Feed forward Neural Network

Component	Main job
Embedding	Turn tokens into vectors
Self-attention	Let tokens interact with one another
FFN	Transform/process each token's representation

Text



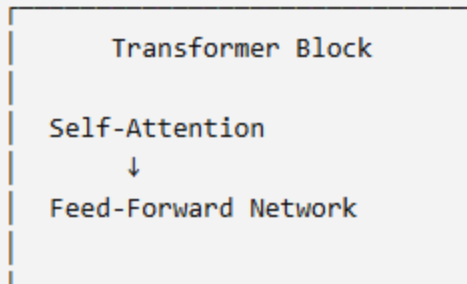
Tokenization



Token IDs



Embeddings + positional information



Repeat many times



Contextual representations



Prediction of next token / output